ActInf GuestStream 053.1 ~ A case for chaos theory inclusion in neuropsy

GuestStream | A case for chaos theory inclusion in neuropsychoanalytic modeling | 1:26:39

Hello and welcome everyone. This is guest stream number 53.1 at the Active Inference Institute.

It's August 23rd, 2023. We're going to be having a presentation followed by a discussion

around the recent work of Alexei Toczynski, a case for chaos theory inclusion in neuro-psychoanalytic

modeling. So if you're watching live, please feel free to add any questions into the live chat.

Otherwise, we will now have the presentation followed by a discussion with our excellent

panelists. So thank you everybody for joining and Alexei to you for the presentation.

Thank you, Daniel. Thank you, Mark, for finding the time. I'm very grateful and excited to talk

about this. Thank you, Andrea, Bryn, Albert. I hope maybe Mike will join us. And thank you,

Daniel and Active Inference Institute for organizing this meeting. So can you see the slides?

Yep. Okay. So I'll just proceed then. This is the outline of the talk. I'll share some preliminary

comments and talk briefly about assumptions in the current models, summarize the data, talk about free

energy principle formulations as compatible with chaos, and talk about some future developments.

And hopefully we can discuss that. I'm very much looking forward to have more questions than answers.

I'd like to start with what Michael Levin taught me, I guess, is just why it's important.

To me, the most important thing is the clinical applications. I think that with your work,

Mark, and with Yuck Panksepp's work, we have been moving away from dualism over the course of some time.

But I think the other big D, determinism, is still highly prevalent and dominant. And I do think it

influences our clinical work, diagnosis and psychotherapy and assessments. And I think that it may be

beneficial for some of the harder conditions we work with to consider including nonlinear framework.

That's my main focus. This is the link to the preprint and the paper is published fully in

NeuroPsychoAnalysis Journal. Some of it is a discussion with this foundational paper that you, Mark, wrote,

the new project for a scientific psychology general scheme, which is a perhaps revision,

a revival of Freud's famous paper, Project for Scientific Psychology. And I'd like to give you a

credit mark that you have been opening conversations between disciplines that hadn't been talking to

each other prior to your efforts, such as psychoanalysis and neurosciences. You've been doing it for more than

20 years with great success. In the same spirit, I wanted to mention that Society for Chaos Theory in

Psychology exists since 91, which is when their first conference happened. About less than 20 years,

then the first conference on Chaos in Lake Italy, around Lake Como in Italy in 76, which was mostly physicists at that time.

And then psychologists joined. And I've attended the recent conference in Toronto, which was absolutely wonderful.

And I've met extraordinary clinicians and scientists. And I think that it would be beneficial to open that dialogue as well,

and maybe if there is a wall between us to consider demolishing the wall. In terms of intent for the paper, you, Mark, pointed out to me that you've had an exchange with Robert Galitzer-Levy. For those of you who may not know him,

he's a psychoanalyst from Chicago, a psychoanalytic writer and teacher who wrote a book on nonlinear psychoanalysis,

and deserves a lot of credit for his advocacy. And, you know, he's sharing his thoughts. He posed a question to you, Mark, of possibly considering nonlinear components in that seminal paper, your project for scientific psychology. And to which he responded that the key point in such discussion was whether the mental apparatus was indeed nonlinear.

Mark Giles- So I took your comment as a task, as a challenge, if you wish, and decided to write a paper about it. I don't know if the data presented is sufficient, but I tried.

Mark Giles- And very clearly, I don't intend to contradict or criticize, you know, that would be funny of me to do. I am hopeful maybe to add a branch on this tree that you're growing, clinical and theoretical tree. And should you consider it worthwhile to go in this direction,

Mark Giles- We wouldn't have to change very much, because this is the paper by Fristen, Lancelot, Dacosta, and others called Stochastic Chaos and Marker Blankets.

Mark Giles- Free Energy Principle is compatible with the chaotic attracting set. What I thought was not present just yet was a formulation where the attracting set experiences a phase transition.

Mark Giles- And I took liberty to ask Carl about it. And he shared some very nice thoughts that I will, with his permission, share at the end of the talk.

Mark Giles- I'd like to cite Maxwell Ramsted, whom you know, and his colleagues from the Papern-Basin, Mechanics of Physics, Often by Beliefs, where they talk about dynamics, mechanics, and principles.

Mark Giles- And in physics, you know, Kepler's work is descriptive. It shows, you know, what the orbits of heavenly bodies are.

Mark Giles- Mechanics are the how of the dynamics. This is Newton's laws. And principles are prescriptive.

Mark Giles- They are about why things work this way. There's a certain hierarchy in physics where you could derive the second law of Newton from principle of least action.

Mark Giles- And then if you have initial conditions, you could calculate the dynamics.

Mark Giles- The reason I'm mentioning that is, I think, Mark, your new project for scientific psychology is a high road.

Mark Giles- It seems to start from first principles, such as the free energy principle, and then you seem to derive things down from there.

Mark Giles- So I thought, why not try to see how the low road may look like?

Mark Giles- If in addition to these derivations and deductions and declaration of what entropy should be in the brain, what if we measure it in animal studies and human studies and health and pathology and see what the data shows?

Mark Giles- And whether or not the low road meets the high road in the middle, or perhaps something may need to be added.

Mark Giles- I also like to clarify why I spend time in the paper talking about obvious things for everybody here.

Mark Giles- The axioms and postulates.

Mark Giles- And I'm quoting here, Edward Frankel, who is a mathematician at Berkeley.

Mark Giles- And you know, Michael will forgive me, but you know, a viewpoint is that one can say that some parts of biology can be seen as based on chemistry, which can be seen as based on physics, which can be seen as based on mathematics.

Mark Giles- So how do things work in mathematics?

Mark Giles- We know that Carl Fristen likes to translate nearly everything he works on into the language of mathematics.

Mark Giles- Classically, there's something called a formal system, which we can metaphorically compare to building a house.

Mark Giles- So the foundation of the house is a set of axioms that has certain qualities.

Mark Giles- And then we build things up using the logic of Aristotle, and we derive things like theorems.

Mark Giles- An example would be Euclidean geometry, which is on the plane, that is based on five axioms of Euclid.

Mark Giles- That's the foundation.

Mark Giles- And then you could derive things like Pythagorean theorem.

Mark Giles- Now, if we were to change just one of the axioms in the foundation, the entire house changes.

Mark Giles- An example would be the fifth axiom of Euclid, that parallel lines essentially don't intersect.

Mark Giles- If you change that, then we could have a house on the sphere, which is Lobachevsky-Riemann geometry, and the radians on the earth are parallel and they intersect.

Mark Giles- Or you could have parabolic shape, where parallel lines also intersect.

Mark Giles- So we'll have an entirely different framework.

Mark Giles- And it may be naive to expect the theorem to look the same.

Mark Giles- The theorem, if we put it on the sphere, is different.

Mark Giles- The formula is different.

Mark Giles- I'm mentioning that is because I love multidisciplinarity and I think that's the way of the future.

Mark Giles- With that, sometimes when we cross over into another discipline, such as psychologists, you know, taking mathematicals or physics ideas, then we may occasionally forget this formal system thing.

Mark Giles- For example, when we say things like human beings resist or violate the second law of thermodynamics,

Mark Giles- Then we, the second law was formulated for closed systems in the state of equilibrium, such as a sealed chamber of gas.

Mark Giles- And human beings are neither closed nor in the state of equilibrium.

Mark Giles- So it doesn't quite apply.

Mark Giles- It's a different formal system.

Mark Giles- One needs physics of open systems to make a statement like that.

Mark Giles- So again, in mathematics, because you, Mark, seem to have taken us deeply in the main of mathematics in your project,

Mark Giles- You have differential equations for action, perception, precision.

Mark Giles- You have, you know, Mark of blankets, random dynamical systems, you know, equivalences, partial derivatives.

Mark Giles- Then in mathematics, people fix the set of axioms.

Mark Giles- They fix the definitions and then they build a theory, right?

Mark Giles- So I want to be clear.

Mark Giles- I want to clarify what are the axioms in the theory that is built?

Mark Giles- What is the foundation?

Mark Giles- And in the effort of time, maybe I'll talk about just one of them, which is Freud's psychic determinism.

Mark Giles- And this is a question for me, not an answer.

Mark Giles- I wanted to ask you because I am assuming whether you kept this axiom or you changed it.

Mark Giles- So Freud said in the psychopathology of everyday life that there is nothing arbitrary or undetermined in the psychic life.

Mark Giles- There's different versions of the statement and it has been repeated by generations of psychoanalysts over 100 years and it became very entrenched.

Mark Giles- It's sort of a cultural meme and a major tenant of psychoanalysis.

Mark Giles- If I were to give an illustration, perhaps, those of us who are not in psychoanalytic therapy, then let's imagine an analytic therapist hearing a patient who intends to say same with an S, but they use the word shame.

Mark Giles- So such a phenomenon, first of all, I think psychodynamic psychologic therapist would probably say that's not an accident.

Mark Giles- Accident is not a used term in psychoanalysis.

Mark Giles- And secondly, they will probably infer of what it might mean.

Mark Giles- For example, infer along the lines of unconscious or emotional dynamics, emotional conflict.

Mark Giles- Somebody else may infer that it may have neuro psychological reasons.

Mark Giles- Somebody else may infer that it probably is a migraine.

Mark Giles- But what you might will not hear in psychodynamic culture is that that was an accident, that it was due to chance.

Mark Giles- I think that chance, accidents, randomness, stochasticity, chaos are not used, they're bad form in psychoanalysis.

Mark Giles- And that is the illustration of, I think, for its assumption, ultimately it's an axiom, it's a belief that is not questioned.

Mark Giles- If we were to translate that assumption in the Friston language, this is the Langevin equation from which Friston starts, which is, has notations by Daniel.

Mark Giles- Thank you, Daniel.

Mark Giles- And we have the change in the state of x equals the flow plus the omega.

Mark Giles- Omega is the stochastic component.

Mark Giles- It's the, you know, in Daniel's metaphor, we have a wave and then we have ripple.

Mark Giles- So we would have to change the stochastic component if we were to go with the Freud's axiom.

Mark Giles- And then if we do that, we would find ourselves squarely in the domain of cold physics, Newtonian physics, Laplace's demon.

Mark Giles- Laplace famously said, give me a position and momentum of every particle and I'll tell you exactly where they have been before and exactly where they will be in the future.

Mark Giles- Now, Freud didn't make statements like that.

Mark Giles- But this is what would happen if you kill the stochastic component.

Mark Giles- I want to be clear that it's not for its fault at all.

Mark Giles- Well, first of all, you need axioms to build a theory.

Mark Giles- He chose that axiom.

Mark Giles- That's the theory he built.

Mark Giles- Second of all, it wasn't exactly mathematics.

Mark Giles- And finally, the zeitgeist of the time was deterministic.

Mark Giles- Perhaps there were initial derivations of stochastic mechanics and quantum mechanics.

Mark Giles- But Freud, when he was learning, the world of deterministic physics was dominant.

Mark Giles- And there's an important nuance in this discussion pointed out to me by the first peer reviewer of the paper,

Mark Giles- who said, you know, listen, if you have something that changes like a stochastic thing,

Mark Giles- and then you apply statistics to it, such as variance, then the variance of a changing variable will be deterministic.

Mark Giles- And then if we look at things like words, which are used a lot in talk therapy, like apple,

Mark Giles- and apple is a semantic memory, it's a category.

Mark Giles- It's a course grading already from green apples and red apples and all other sub kinds of apples.

Mark Giles- So in that regard, words are deterministic things.

Mark Giles- So perhaps when we have a dream analysis session and the patient shares with an analyst,

Mark Giles- You know, the movie that they saw, the messy, you know, irrational thing with images and tunes and anything,

Mark Giles- and maybe words even, if they narrate it, then they slow things down, they average things out,

Mark Giles- and they put it in deterministic tools, which is words.

Mark Giles- So with some objects, we can apply deterministic tools.

Mark Giles- But I want to point out that the foundation that everything and anything in the psychic life,

Mark Giles- that all the mentations have known stochasticity randomness is a very strong assumption.

Mark Giles- And I'm curious what would happen if we were to allow ourselves to see things on the sphere or in the parabolic shape.

Mark Giles- It's not a bad exercise because the general theory of relativity would not have existed between without these other geometries.

Mark Giles- In terms of definitions, this is a familiar formula that you Mark use of Shannon's entropy.

Mark Giles- He originally wrote about telegraph and a sender and receiver in a fixed alphabet.

Mark Giles- But here, if we draw an attracting set, which happens to be Lorenzo Tractor in phase space,

Mark Giles- and we look at these, I don't know if you can see them, they're small, the white points, you know,

Mark Giles- that's the initial condition.

Mark Giles- And then they spread out and evolve as time goes by.

Mark Giles- But you can see the density of the attracting set.

Mark Giles- It is higher density and lower density.

Mark Giles- So we could calculate probability distribution.

Mark Giles- A slightly different question is this.

Mark Giles- What would happen if you were to change the initial condition a little bit,

Mark Giles- one millimeter in phase space?

Mark Giles- Different things can happen and one of them is shown in this diagram,

Mark Giles- that would be divergence.

Mark Giles- And the formula here is initial condition multiplied by exponent of lambda t.

Mark Giles- So when lambda is positive, you have divergence.

Mark Giles- And it's called the exponent of exponent and that's one of the criteria for chaos.

Mark Giles- When it is negative, we have convergence and when it is zero, we have equidistant trajectories,

Mark Giles- we have Newtonian mechanics, right?

Mark Giles- This is an illustration of a chaotic system, which is a great red spot on Jupiter.

Mark Giles- And it fits perfectly the Fristin's paper about the Markov blankets and stochastic chaos,

Mark Giles- because that thing is stable.

Mark Giles- We have been observing it for 300 years.

Mark Giles- One could say that this elliptical shape is the Markov blanket around the chaos that maintains its integrity.

Mark Giles- It's not a contradiction because in one spatial scale, you have Markov blanket and on another spatial scale,

Mark Giles- you have a chaos within Markov blanket.

Mark Giles- This is a convergent system that is a bit trivial to all of you.

Mark Giles- This is a hole in the ground and then surrounded by hills on all sides.

Mark Giles- And if you were to put a ball on the hill, it's going to roll down in the potential well.

Mark Giles- And typically in phase space, we'll call the bottom of the hole an attractor and we'll call the hills repeller.

Mark Giles- Repeller is a term less frequently used, but you know, it's the highly unstable point in phase space from which things move away from and they move toward the attractor.

Mark Giles- Now that's just one attractor.

Mark Giles- The interesting things start happening if you've got two of them.

Mark Giles- Imagine a second hole immediately to the right.

Mark Giles- And the boundary between them may very quickly become chaotic.

Mark Giles- Here's another illustration on a convergent system from the active inference textbook that I hadn't read when I was working on the paper.

Mark Giles- I discovered it later.

Mark Giles- The chart on the left is the density.

Mark Giles- And again, we see high density in the center and we see lower density in the periphery.

Mark Giles- And the chart on the right is surprise.

Mark Giles- We're very not surprised to find a ball in the center and we're very much surprised if it stays on the hill.

Mark Giles- And I'll read you a citation.

Mark Giles- Active inference is restricted to systems like that one on the left, which count to random fluctuations with their average flow,

Mark Giles- And thereby retain their form over time.

Mark Giles- It may sound like this statement contradicts what I said earlier, that FEP is compatible with chaos.

Mark Giles- But it doesn't because you could have different spatial scales and convergence on one scale and divergence on another.

Mark Giles- I also historically think that we have been staying converging systems mostly so far.

Mark Giles- Now, if we move from things like that, which is very simple, to a mind or consciousness,

Mark Giles- Then I think that what we could be looking at is this.

Mark Giles- That's a landscape with attractors and repellers.

Mark Giles- That is very complex.

Mark Giles- And I'm going to quote the former president of the chaos society for psychology is that that landscape is dynamic.

Mark Giles- It moves.

Mark Giles- Perhaps some things on it are stable.

Mark Giles- If we were imagine a clinical scenario where there's a patient who is very prone to shame.

Mark Giles- And very averse to express anger.

Mark Giles- Then we could say that shame is an attractor and anger is a repeller.

Mark Giles- But we work with pattern things in dynamic therapy, so they could be relatively stable until the work in therapy makes them more shallow distributions.

Mark Giles- But certain things change overnight.

Mark Giles- Some things change in a week or during the course of therapy.

Mark Giles- So I mean, strictly speaking, that's a dynamic landscape.

Mark Giles- And I don't think we model things this way.

Mark Giles- I'd like to also quote Maxwell Ramstad from the same page.

Mark Giles- If we have a paper that free energy principle is a tale of two densities, it would be incomplete to say that we just minimize one variable.

Mark Giles- You know, what happens is we minimize the Shannon's entropy of probability distribution over sensor states, while at the same time we maximize entropy of internal beliefs, which is very important.

Mark Giles- That's adaptation.

Mark Giles- If somebody has rigid internal beliefs with sharp probabilities and then COVID happens, they will have lower probability of survival.

Mark Giles- They're not adaptable.

Mark Giles- So adaptability means maximization of entropy of internal beliefs, while maintaining the form means the minimization of Shannon's entropy of probability distribution over sensory states.

Mark Giles- Here's another term called Magorov-Sinai entropy that is not intuitive and it's from topology.

Mark Giles- The picture you see is the photograph of the statue called Intuition outside the Fields Institute of

Mathematics in Toronto.

Mark Giles- Here's a graphical illustration.

Mark Giles- If you were to draw the trajectories in phase space and surround it by certain volume, say a circle with a radius epsilon, and ask yourself what happens to this volume as they evolve?

Mark Giles- Well, when it expands, then you can have chaoticity.

Mark Giles- When it becomes more compact, you can have convergence.

Mark Giles- When it stays the same, you have periodic system, which is indicated on the chart on the right, where you have a correlate of the Kolmogorov-Sinai entropy on the y-axis and you have this epsilon on the x-axis.

Mark Giles- So the flat line at the bottom where the circle doesn't change is the periodic system.

Mark Giles- The curved line in the center that converges on a k is the proper chaos.

Mark Giles- This is theoretical chaos.

Mark Giles- And the upshoots from it is what Friston calls stochastic chaos.

Mark Giles- This is chaos plus noise.

Mark Giles- And then you can have pure noise and then you can have pure linearity.

Mark Giles- So all of these things exist.

Mark Giles- They're all on the table.

Mark Giles- You can have geometry on the plane, geometry on the sphere, geometry in the parabolic shape.

Mark Giles- So what does the data show?

Mark Giles- And when we look at EEG entropy, people use approximate entropy, permutation entropy, and spectrum entropy, which are all derivations of Kolmogorov-Sinai.

Mark Giles- And the first two are in different phase space from the last one.

Mark Giles- I believe Mark, correct me if I'm wrong, you cite Geraseris study in your book, Hidden Spring, when you talk about entropy and phases.

Mark Giles- And they use spectrum entropy, which is a derivation of Kolmogorov-Sinai in the frequency domain analysis phase space.

Mark Giles- So the data shows that systems for as small as the axon of a squid isolated in the lab to single neurons, to coupled neurons, to EEG, we have evidence of positive, not going to be up enough exponents in some regimes of operation.

Mark Giles- With EEG, we can be more specific that alpha-rhythm seems to be weakly nonlinear, while things below alpha, like delta, seem to be more periodic with noise, and things above alpha, such as gamma and high gamma, seem to be chaotic.

Mark Giles- So if we were to summarize this data, then it would be, we can say that Kolmogorov-Sinai entropy appears to increase as generalized arousal increases from delta to gamma rhythm.

Mark Giles- So if we were to summarize, Kolmogorov-Sinai entropy seems to be higher in healthy alert brain mind functioning than in the states of coma, seizure, deep sleep, and it can stay close to zero in a coma or deep sleep.

Mark Giles- The interesting anecdote here is that a simple act of closing your eyes when you're alert decreases Kolmogorov-Sinai entropy.

Mark Giles- Clinical applications are vast and developing fast, for example, except for clinical psychology, I must say, and psychiatry and psychotherapy.

Mark Giles- But in neurology, the seizure detection on the EEG is right now can be done with AI with about the same accuracy as a human neurologist.

Mark Giles- And that algorithm does use chaoticity assessment.

Mark Giles- There's models for Parkinson's, neuropathic pain, and I have other examples that I'll skip for right now.

Mark Giles- This is a graphic illustration of phases of sleep.

Mark Giles- And again, Kolmogorov-Sinai entropy as a phase of sleep.

Mark Giles- It shows wakefulness and then shallow sleep, progressively deeper sleep, and REM.

Mark Giles- And that's the time evolution.

Mark Giles- If you move REM in frequency to where it belongs to wakefulness, then you'll see the same progression that Kolmogorov-Sinai entropy appears to increase as a function of generalized arousal.

Mark Giles- This is a seizure onset at about second four.

Mark Giles- Seizure is a reduction in chaoticity.

Mark Giles- Healthy functioning is higher chaoticity than a seizure.

Mark Giles- This is a coma, which is relatively invariant theta and delta frequency.

Mark Giles- This is a nearly periodic system with some noise.

Mark Giles- And death is a linear system.

Mark Giles- It doesn't change.

Mark Giles- This is persistent vegetative state, minimally conscious state, where researchers were asking a question on prognosis of survival.

Mark Giles- And they've concluded that diversity and variability of the EEG was the predictor of survival.

Mark Giles- Higher entropy means life.

Mark Giles- So the hypothesis is that chaotic, stochastic, and linear processes as well as hybrid ones, such as primarily chaotic functioning with noise, can be present concurrently at different scales of the brain mind, or the same scale but in different places, or at different times.

Mark Giles- I realize it's a mouthful, but it fits the data presented in the paper.

Mark Giles- This is again a picture from the textbook, Enough to Your Inference, by Thomas Parr, Carl Fristen, and Jovan K.

Mark Giles- It's a cortical column.

Mark Giles- And the errors you see here, ascending and descending connections.

Mark Giles- If by chance you're not familiar with the framework, they correspond to the ascending errors are prediction errors and descending ones are predictions.

Mark Giles- This is what they say in the book.

Mark Giles- But there's an asymmetry in message passing.

Mark Giles- And the reason for that is that the operations required to compute prediction error from expectations are nonlinear.

Mark Giles- This nonlinearity is due to computation of predictions using nonlinear functions that tend to increase the frequency of the signal.

Mark Giles- I'd like to talk a little bit about reduction.

Mark Giles- It became very fashionable and sometimes silly to, you know, use this like, oh, your reductionist, now your reductionist.

Mark Giles- The second most popular term is oversimplify, you oversimplify, now you oversimplify.

Mark Giles- To me, they're not bad words.

Mark Giles- They're necessary tools to simplify and to reduce.

Mark Giles- You know, the map that equals the territory exactly is useless.

Mark Giles- Nobody's going to use it.

Mark Giles- It must kind of, I'm quoting James Glick, simplify and abstract.

Mark Giles- But, you know, I'll illustrate the specific term.

Mark Giles- If the car is moving with a fixed velocity on a straight line, and if I know exactly where it is right now, I can tell with certainty where it has been an hour before.

Mark Giles- That is a reduction in time.

Mark Giles- And I can do that because the formula is linear.

Mark Giles- It's distance equals velocity multiplied by time and velocity is constant.

Mark Giles- If the watch is broken and I disassemble it and I find a faulty part, and then I put it back together, and it works, then it's a reduction in space.

Mark Giles- And I can do that because watch is not an end colony, to use Daniel's terms.

Mark Giles- Watch is equal to the collection of its parts.

Mark Giles- So I wanted to talk about, Mark, your mention of the predictive hierarchy composed of billions of homeostats in your work.

Mark Giles- This is an approximate picture.

Mark Giles- This is not the exact picture of what you talk about in the new project, and this is from your hidden spring.

Mark Giles- It's a predictive hierarchy where predictions flow from left to right and prediction errors from left to right.

Mark Giles- And one of the statements in the book, in the hidden spring, you say that brains many complex functions really can ultimately be reduced to a few simple mechanisms like this.

Mark Giles- So it appears that you, you know, you call it reduction.

Mark Giles- But I just wanted, it is a powerful model.

Mark Giles- It is a flexible model.

Mark Giles- I wanted to describe some features of it.

Mark Giles- Well, first of all, if homeostat is present at every single scale, then we have this homogeneity of mechanism throughout the scales.

Mark Giles- And I want to say that in physics, things don't look this way.

Mark Giles- Quantum mechanics is the microphysics.

Mark Giles- Statistical mechanics is molecular level.

Mark Giles- Newtonian mechanics is heavenly bodies.

Mark Giles- And then we have general theory of relativity.

Mark Giles- There's qualitative shifts.

Mark Giles- There isn't a homogeneity of mechanism all the way throughout.

Mark Giles- In fact, when people say, take entanglement and they apply to human beings, that's sort of a misunderstanding of entanglement.

Mark Giles- The second comment is from Michael Levin, who talks about the scales.

Mark Giles- And if you know, his comment is that you could have different functional spaces at every scale, where you could solve the genetic task on one scale, electrochemical on another, morphological on the third one.

Mark Giles- And then how you do message passing, what is the common language is not a trivial question.

Mark Giles- Along the same lines, I've already mentioned that a message between the scales can be nonlinear if you do prediction errors.

Mark Giles- Now, if you take one nonlinear message and then you multiply it by billions, you're going to have a full on turbulence, which is a chaotic system.

Mark Giles- It's a dynamical system.

Mark Giles- And turbulence, you cannot reduce to homeostat or anything else.

Mark Giles- Saying that water consists of H2O does not explain turbulence.

Mark Giles- So I don't know, that's an open question, but I wanted to maybe share some perceptions of this model.

Mark Giles- This I talked about, and this is the Fristen's thought about phase transition in the attracting set.

Mark Giles- He said that if you have a hierarchy of Lorenzo tractors, a slower one and a faster one, and a slower one governing the transitions of the faster Lorenzo tractor,

Mark Giles- And then you could have a Feb formulation with attracting set experiencing changes and transitions.

Mark Giles- And that's brilliant, I think, and it matches the anatomy because the brainstem and the subcortical structures appear to be slower Lorenzo tractor that could be seen as governing the transitions of the cortex, which is astral Lorenzo tractor.

Mark Giles- I know I'm over-sup with having things a great deal, but it sort of, this is his thoughts.

Mark Giles- I think him and Lancelot de Costa are working on that.

Mark Giles- Perhaps Maxwell Ramstad's group is working on that as well.

Mark Giles- The clinical applications, the landscape that I've mentioned before would be absolutely lovely to see if it were to consider going there.

Mark Giles- Borderline personality disorder, or another term for that could be complex trauma.

Mark Giles- The metaphor that we use for these patients is they're predictably unpredictable.

Mark Giles- That the only thing we know on Wednesday that we don't know what's going to come up, suggesting chaoticity.

Mark Giles- Acute trauma has a symptom of hyperarousal, suggesting possibly higher level of chaoticity.

Mark Giles- An onset of mania can be seen as a transition into a higher energy state.

Mark Giles- And addictions, particularly with stimulants, can be seen as phase transitions into a chaotic state.

Mark Giles- It may sound like what I said now is contradictory to what I said before, when I suggested that, you know, lower chaoticity could be a problem, such as an onset of a seizure.

Mark Giles- And here I'm saying higher chaoticity can be a problem.

Mark Giles- But that would only be a contradiction in the linear world.

Mark Giles- In the real world where linear is very small percentage of what's happening.

Mark Giles- I'm saying both that too little and too much chaos could be a problem.

Mark Giles- Just enough chaos imbalance could be perfectly fine.

Mark Giles- This is a graphical illustration of what can be done.

Mark Giles- This is from my colleague and friend David Pinkus, who is a professor in Chapman University and I think also former president of the Society for Chaos and Psychology Society.

Mark Giles- These are trees and may sound like completely unrelated to what we do, except if you look at a human lung or dendrites in a neuron.

Mark Giles- And you can see a healthy tree on the right from me and a nearly dead tree on the left.

Mark Giles- And if you do the fractal assessment, fractal dimension assessment on them by counting the thickness of branches and count the number of branches at each level,

Mark Giles- Then you can quantify it and then you can have approximate quantification of health and resilience versus in pathology.

Mark Giles- And again, as exotic as it may sound, this is very similar to what people do when they detect a seizure on the AGE.

Mark Giles- They measure CD correlational dimension, which is approximately fractal dimension.

Mark Giles- And I think my 30 minutes are up.

Mark Giles- Thank you so much for your attention.

Mark Giles- Thank you, Lexi.

Mark Giles- Great presentation.

Mark Giles- Well, looking forward to this discussion.

Mark Giles- First, Mark, please feel free to add your reflections.

Mark Giles- Then, Andrea, then we'll continue on.

Andrea Andrea- Thanks very much.

Andrea- When Alexi asked me if I would join this panel, the first thing that I said to him was yes, coupled with the second thing being,

Andrea Andrea- But please be aware that I know next to nothing about chaos theory.

Andrea Andrea- So, you know, I'm not sure how much of a contribution I can make to a technical discussion of this kind.

Andrea Andrea- So, I'm just repeating that before I say anything else.

Andrea Andrea- If I say anything that sounds amateur, that sounds as if this guy doesn't know what he's talking about when it comes to chaos theory, it's because I'm an amateur and I don't know what I'm talking about when it comes to chaos theory.

Andrea Andrea- That said, the second thing I want to say is I'm not entirely sure what is at stake here.

Andrea Andrea- You know, it seems as if Alexi feels that it's really important that I and our colleagues get our head around chaos theory and bring some of its, what it has to offer into our field.

Andrea Andrea- Because I don't know much about it, I don't know what's at stake, you see, I don't know why that's important.

Andrea Andrea- So, you know, perhaps somewhere along the line in the next just less than an hour that we have, perhaps Alexi, you can clarify that for us.

Andrea Andrea- The third thing I wanted to say is, and perhaps in order to illustrate the first thing, is that as Alexi was a leader,

Andrea Andrea- As Alexi was talking now, it seemed as if he was talking about chaos, as if it was synonymous with stochasticity, or, or randomness.

Andrea Andrea- And that's not my understanding.

Andrea Andrea- My very limited understanding of chaos theory was that it's not a synonymous chaos as used in that, in that science or in that theory is not synonymous with with randomness.

Andrea Andrea- It's rather synonymous with unpredictability.

Andrea Andrea- Because we don't know the initial conditions of the system or we can't specify all the variables with sufficient precision in their initial conditions, it becomes impossible to predict.

Andrea Andrea- And because of the nature of the interrelationships of the variables one tiny little error in your estimation of the initial conditions can lead to massive consequences.

Andrea Andrea- And this is why it's so hard to predict chaotic systems.

Andrea Andrea- So if that is what is meant, then I would say, sure, of course, this applies to the way that the mind works and the way that the brain works.

Andrea Andrea- But that leads me back to my second point.

Andrea Andrea- I'm not sure what's at stake.

Andrea Andrea- But I would say if that's what you're asking, if that's what you're claiming rather, Alexi, then I would say that that surely must be so.

Andrea Andrea- That it is exceedingly difficult to predict mental and neural events at any level of complexity.

Andrea Andrea- Now, I think I've said three things.

Andrea Andrea- Let me now say a fourth thing.

Andrea Andrea- I think that a crucial.

Andrea Andrea- Well, let me put it this way.

Andrea Andrea- Of course, what the brain and mind are trying to do.

Andrea Andrea- It's a matter of trying to predict what's going to happen.

Andrea Andrea- So when I what I've just said about how how difficult it is to predict what's going to happen in the brain or mind.

Andrea Andrea- It's it's it's it's almost the same thing as to say that the mind and brain are trying to model the world.

Andrea Andrea- That's that's the main thing that we're trying to do is is is is have a good model of the world so that we can get some sort of grip on this chaos.

Andrea Andrea- So that we can act rationally in the sense that we can act in order to preserve ourselves that we do not dissipate.

Andrea Andrea Andrea- By just exploring all possible states and and no longer existing as a system, because we as systems need to remain within certain bounds across a great many variables, as you said.

Andrea Andrea- We can't afford to just let it happen and you know and throw ourselves to it's to to to its face.

Andrea Andrea- We need to try to predict in a very unpredictable world.

Andrea Andrea- What must I do in order to meet this need, given how the world works.

Andrea Andrea- So the brain is trying to model the world, our predictive model is is is is trying to synchronize itself with the world and for that reason it's going to be very chaotic.

Andrea Andrea- But the crucial point is it's trying to reduce the unpredictability.

Andrea Andrea- With that said, I'll make a fifth point in or maybe it's another part of my fourth point.

Andrea Andrea- No, I think it's a fifth point, which is that the way that I conceptualize the predictive model.

Andrea Andrea- And it's not me alone who does this is that it is a hierarchical model.

Andrea Andrea- And I think that's very important and perhaps here's where things do start where it starts to.

Andrea Andrea- At least in my mind, I start to see some of what's at stake.

Andrea Andrea- I think that the more peripheral layers of the predictive model, I see the predictive model as a as a as not as a triangle, but rather as an onion.

Andrea Andrea- So the core is the the deepest layers of the model, that is to say, the the layers that have the the greatest.

Andrea Andrea- As a spatial and temporal generalize ability.

Andrea Andrea- And they're the most stable layers of the model.

Andrea Andrea- And as we move toward the periphery, which coincides with the periphery of the of the nervous system, in other words, the encounter with the outside world.

Andrea Andrea- So it becomes more chaotic, we are able to talk we can tolerate more unpredictability at the periphery.

Andrea Andrea- And we were trying to and that's and that also allows for greater accuracy, because the world is so unpredictable.

Andrea Andrea- We have a more more chaotic.

Andrea Andrea- Outer layer of our predictive model, where things also there's less that's at stake, you know, is that car going to turn left or right?

Andrea Andrea- Well, I thought it was going to turn left turn right.

Andrea Andrea- So what doesn't matter, you know, but it is the law gang to is the car gang to obey the laws of gravity that matters more.

Andrea Andrea- You know, that's not something that that's a much more consequential surprise if it if it does not, and perhaps that's a silly analogy, but at least it illustrates the general point.

Andrea Andrea- Our most heartfelt beliefs and that is to say, the deepest layers of our of our predictive hierarchy there, we find the same patterns over and again, and we, we find ourselves.

Andrea Andrea- We are more certain of our predictions and we act more stereotypically in accordance with them, whereas at the periphery that does not apply.

Andrea Andrea- So I think at one point in your presentation, Alexei, you were saying something like, you know, one size doesn't fit all it's not all just one mechanism all the way down.

Andrea Andrea- Of course, I completely agree with that.

Andrea Andrea- The the the it's not just one mechanism at the different homeostats that you were referring to on their different scales, also serving different needs, you know, and there's there's not a common currency.

Andrea Andrea- I think that's a very important part of of how consciousness works, why we have quality or at all, is because we can't reduce things to a common currency we're dealing with categorical variables.

Andrea Andrea- At the very least in relation to our bodily needs, they are categorically distinct needs and they need to be treated as such.

Andrea Andrea- And this is why we have different feelings in relation to these different categories of of free energy.

Andrea Andrea- So those are those are some of the main things I wanted to say just to get the ball rolling.

Andrea Andrea Andrea- mindful of the fact that Andrea Kelly

Can you hear me now? Yes, sorry. I will try also to articulate something of the very fascinating paper of Alex Yeben. I must confess, since I'm also a psychoanalytic psychotherapist, a psychiatrist, it's not my field, I'm not a physicist, so I can really say a few words. Some is of, as I said already, fascination, because in general, I hate to stick to rigid theories. So if there is a paper like Alexei who can chaoticize my theories, even in psychoanalysis, in the workings of the mind and the brain, I'm happy to dive into it. At the same time, I don't have the skills to discuss the math part of his papers or even the physical formulas and so on. But I was reminded of a very early, and I think you all know that book, Jacques Monod's Chance and Necessity, which is a classic. And I think it's a book in which you can even from scratch. I mean, here that our aim in the world, and I know I'm sounding theological, but we have theological aims. In order to stay alive, we have to face chance and chaos. So I think this is, in order to survive and reproduce, we have to have a mathematical system to reduce chaos and to bring up again determinism or psychic determinism. Since one of the points that Alexei, I think, raised is, can we still hold the point of Freud's about psychic determinism? And I think if you apply not to the physics of just materialism of things actually, but to the living organism, I think we do. This is my point and I would like to hear what Alexei says. In the end, the chaos theory, mathematics, as far as I understand, it must be a mathematics who reduce chance or chaos to a linearity, at least in the living organism.

And I wonder if and what Alexei thinks about it, because you seem to disagree with that.

Want to give a direct response, Alexei? Or then Bryn and Albert would be awesome to hear your takes as well.

Some of the thoughts from Mark and from Andrea.

Should I start by responding to Andrea's points or Mark's?
Maybe I'll start by, because I think the key point that Mark, you brought up is what's at stake is still unclear.
My impression is, and please correct me if I'm wrong, I was very influenced, Mark, by your paper,
I forgot exactly the journal, where you talked about psychiatry world, how these sort of, you know,
convention based diagnosis of here's a major depressive disorder, here's a bipolar disorder,
nothing of which has to do with etiology, you know, convention and statistics, you know.
And then you brought up this, you know, revolutionary thing with effective neuroscience, where you talked about the manic state and the depression state, and, you know, tied into the punk subsystem, I think is extraordinary.
What I wanted to say is, you know, if we look at how we diagnose and think about what's happening,
and we look at the language of how we do it, it is still a deterministic language at the end of the day.
So when we moved from DSM-4 to DSM-5, we brilliantly discovered that continuous variables exist.
We used to have buckets, everything was in a separate bucket, and then we said autism is a
spectrum and everybody applauded. But, you know, mathematicians were quietly laughing in the corner,
you know, because of course continuous variables exist, but we don't do dynamics, we're still very static.
Here's this clear-cut thing, and this is the clear-cut continuous thing, right?
And my impression, I may be wrong, is that even in our neuropsychoanalytic developments,
we still do things deterministically. We assess which bucket things belong to,
or maybe two buckets at the same time, or three buckets. But we don't do things like,
you know, how the chaotic systems are qualitatively different from static things.
I'm quoting William Solis. They have transience, they have contextuality, and they have emergence.
We don't do that, you know. And I think that if we embrace the chaos theory, if we do things like
correlational dimension assessment or fractal dimension assessment, starting from this system
of classification of emotions, there's still buckets. Here's a clear-cut thing called fear,
that's a clear-cut thing called rage, buckets, determinism, okay? And, you know,
they look differently in the dynamical systems world. And then the system of psychopathology
could look like, you know, correlational dimension changed, and not that somebody fell into a bucket.
You know, if we look at this complex, you know, landscape of attractors and repellers,
and examine psychodynamic things like, please forgive me if I'm incorrect, but one of the possible
definitions of repression could be seen as not letting things into consciousness. Maybe we can call
it suppression, but what I'm trying to say is, I don't want to think that right now, it hurts,
it's a painful thing. That, in the terminology that I shared, is a repeller. And perhaps when you
work through things in a psychodynamic psychotherapy, perhaps you put a scar tissue on the wound,
it becomes less of a sharp, you know, repeller, and it smooths out. You know, we don't talk,
we don't diagnose like that, we don't treat like that, we don't think like that.
One paper that I'm working on that is in peer review right now, and I can't share too much,
says that, you know, we think like storytellers, I'm quoting Daniel Kahneman, our, our, you know,
thinking mind of a clinician is not a statistician. We don't think in probabilities, we think in labels,

and we think in stories. And I know you have this powerful quote, Mark, from Freud, who said that that's the nature of the mind. And I most respectfully disagree that that's the entire nature of the mind.
I do think that we have things on the mind that are not stories, and not words. So if the nature of the phenomenon is chaotic and stochastic and erotic, then we can use appropriate mechanisms to diagnose and classify that fits the nature of the model. The second thing you said, chaos and stochasticity,
I absolutely agree with you that they're different. You know, on the chart that I've shown, if you converge to a stable, stochastic, chaos is deterministic, forward deterministic. So if initial conditions are changing, they will diverge, but you can predict things forward, not back.
But there is no formula in a stochastic thing. You know, you throw the coin, you don't know, you have no predictability whatsoever, short term or third or long term. So they are different. And in the sense that you're talking to sensitivity to initial conditions, this is music to my heart to hear you, to say that you acknowledge that chaoticity clearly is present in the brain. On which level of the hierarchy in your model is a very important thing. But again, when we draw this picture, whether it's a triangle or an onion with a core and layers, it's a static model. And when you acknowledge that there's higher unpredictability toward the periphery, it could be, but there could be dynamics about the structure of the model. And perhaps what you're saying is survival, you know, like, I must breathe, and I must breathe air, and I'm not in the water. And that is a very stable prediction.
Absolutely. But when we deal with psychopathology, and we deal with things like shame and pride, and awe and other things, then I wonder if we should continue to use this metaphor of a homeostat. I really don't see how homeostat applies to gamma rhythm in the brain or to shame. So a homeostat is the convergent systems, so that you minimize entropy by keeping things in the range. However, if you dive inside the homeostat, if you look at the temperature variation inside our narrow range, you may see chaos, which is exactly the picture that I saw. So that I showed. And to Andrea's point, about the Freud's principle of psychic determinism, yes, I almost think that we haven't said goodbye to that. We haven't tried to say that chance exists in the mental life. Accidents exist. We don't do that. There's a firm tradition in psychoanalytic psychotherapy to say there's no accidents. It's all determined. That's a major key point of the discussion. I don't know if I said too much.
Maybe Andrea, if you want to give any thoughts or how do you want things to continue in the coming minutes in years? Okay, thank you. Thank you, Daniel. Just adding a word that I didn't use because I think it's very important about trying to defend the point about psychic determinism, which I'm very fond of.
I mean, the word is adaptation. I have difficulties in because when you see a psychotic patient, when you see a borderline patient, well, I don't see chaos or total unpredictability. I think adaptation, their disruptions, I try to understand what are the, even the mathematical part behind that, in my opinion, has to have a tendency toward an order which is adaptive. In this way, I see the mathematical part of chaos may be of use. I mean, but bringing to an explanation, not to it, maybe not to an order or to a determinism, but to a point in which there must be this adaptive reason, even in the chaos.
Do you see my point?
I do, but I think there's a deeper issue about how we think, you know, say there's a clinician,

psychoanalytic psychotherapist. We have a working memory and the Miller's law that Mark, you quote
a lot about the seven units of information in the audit to audio verbal working memory. I think in the
Miller's paper, he actually said seven chunks. So I'm sharing something from the paper in review right now.
We cannot think probabilistically because it overloads the working memory. This is why when
I'm a clinician, I think about psychosis, I come up with a story, a formulation that fits in the working
memory. And this influences our work. We don't have the mathematics in here to think in the terms of
probabilistic things and chaos theory and other things. We need tools to do that. And in mainstream
neurology, they use tools. They do talk to the patient that examined the patient, but they have MRI,
they have EEG, they have PASCAN and all the other things. In psychoanalytic psychotherapy, we say,
we don't need any tools. We just talk to a person and that is sufficient for everything we do.
But that's not necessarily true. Beatrice Beebe, who is a psychoanalyst, works with a mother-infant
diet and there's zero words exchanged. But she looks at this movie that she records on a video camera and
then she throws it, shows it in a slow motion. So she takes the flow, which is dynamic flow, not words,
not discrete things. And then she slows it down and she shows the level of attunement between the
mother and the child. We don't do things like that. We talk and we use this deterministic things to
understand the psychopathology of the patients and that's psychosis and that's schizophrenia and that
is bipolar one. This is panic system and this is seeking system. We determine everything. And that is
a limitation. Because inside, I believe, what is a fear cascade? How does fear stop? It's a flow.
It's a turbulent thing. And we put one label on it, say fear. We've classified it. So for some things
it's appropriate, but for some things that may not be it. Maybe, you know, as, you know, Einstein said,
as Mark, you quoted Einstein, as simple as possible, but more simpler than that. So at which point
we are missing with or missing the phenomenon by using deterministic tools in our work.
Thank you, Andreas. Thank you. Thank you, Daniel and Mark to everyone. Thank you. I have to go.
Awesome. So Albert and then Brandon will continue on this topic.
Yeah. Thank you for the presentation and the very interesting discussion so far.
One thing that comes to mind is the, well,
the thing we're talking about right now, the psychic determinism as depicted as being a
foundational law and psychoanalysis. Well, I'm thinking about the authors, the psychoanalytic
authors who came, the prominent ones, who came after Freud, who seem to have
who have room for this chaos. Maybe they didn't use this term explicitly, but I'm thinking,
for example, Wilfred Biong, who's got the notion of the O layer, which has connotations of, well,
the thing that cannot be known or categorized, who has like indeterminate elements of the mind.
And that's one of his major contributions to psychoanalysis. On the other hand, more the French
schools, I'm thinking of Jacques Lacan, who's got the notion of the real, which is also one of his major
contributions to psychoanalysis, where notions of rupture, of absence, or of lack are very prominent.
I'm not sure, I'm not an expert here, but I think Lacan explicitly challenges the notion of Freud's
mechanistic worldview in one of his seminars. So I'm wondering what you think about this, Alexei?
I'm not an expert in Lacan at all. And I know Wilfred Biong in a limited fashion. So perhaps

Marc is a better person to talk. And I like Freud, particularly in Marc's interpretations and
translations, but maybe Marc can comment on Lacan and Biong. I am not sufficiently knowledgeable about their work.

Marc Ben- If you want to, Marc, or Bryn, you can share some thoughts. Yeah, please.

Marc Ben- Well, I think that the point that Albert's making is we must remember that psychoanalysis is not synonymous with Freud, that not everything in psychoanalysis stands or falls by whether or not we accept this or that tenet of Freud's. And in neuropsychoanalysis, we try very much to draw upon all of the traditions within psychoanalysis. And we sort of fondly believe that we'll be able to find more common ground because there are greater empirical tools available in neuropsychoanalysis than in psychoanalysis
by itself. You know, we are hoping to resolve some of these theoretical disagreements, trying to move beyond just different systems of belief, trying to move more toward a psychoanalysis in which we can decide questions in an ordinary scientific way. But it will always be very difficult for exactly the reason that Alexei's whole talk is about, I suppose, which is the extreme complexity and therefore unpredictability of the phenomena that we are dealing with. So I think Albert's points are valid, that the beyonds approach, Lacan's approach, they add a great deal, they don't have as much to lose, if we throw out the principle of psychical determinism. But having said that, let me just quickly say a word about psychical determinism, because this has been a big part of what Andrea picked up on in what Alexei said.

I think it comes down to the question that I asked at the outset. It's not a matter of, is the behavior of the mind random? It's a question of how predictable. It's not a matter of, is there no determinism? It's a matter of how easily can we trace effects back to their causes? So the fact that there's a great deal that goes on where we can't do that. There are many things our patients say, whether they be slips of the tongue or anything else of that kind, and not only to do with parapraxies, but anything that they do. There are certain things that we see with regularity. I mean, the very phenomenon of transference is all about regularities. There are certain things that we see, yep, this I can understand. This happens repeatedly. I think I can explain that with reference to certain aspects of the patients' inner world and their past. But there's a great deal else that we can't. So I don't think that it's a kind of, you know, either there's determinism or there's stochasticity. I think that there's, this is why I was talking also about the layers of the predictive hierarchy, that the deeper layers are the unconscious ones. Their things are pretty stereotyped, pretty predictable, in fact, too predictable. I mean, in other words, it overgeneralizes it. It's too confident about its beliefs. But as we head toward the pre-conscious and conscious peripheries, you know, there things are much less predictable. So that's, I won't say more. There's a couple of things in what Alexei said in his comments after mine, that if we have time, I'll come back to. But I think let's rather hear from from Bryn before I say anything more. Thank you so much. I'm perhaps the least qualified here to comment on your paper, Alexei. But I do have one or two thoughts. And they come from a place of being very fortunate thus far in my life to have been a patient for many, many hours,

doctors, sometimes on the couch, and also to be a clinical psychologist. I failed to mention that at the, in my introduction. So I'm a clinician, I was, until recently, when my foray began into more research-based areas, and by the Active Inference Institute, and Mark's work, and Carl Pristen's work, I was a pure clinician, I had it in my mind that I was going to treat psychopathology to the best of my ability. And that really, it was being a patient, first and foremost, that brought me into closer contact with the theme of your paper, and certainly the themes of Mark's work. And I think I've encountered it from both areas. And it certainly, it matters to my mind, it matters less the inner workings of the degree to which chaos, this is a stochastic, all these big, wonderful terms that I'm only beginning to to even think about. But what matters to me is that, how do you help somebody who is in a state, who comes in disheveled, and, you know, how does one go about helping, aiding somebody in that state? And I fell out of love with psychoanalysis, for personal reasons. But it was, in returning to it, I've become more kind of a bit acquainted with its strengths and its limitations. And I'm realizing that it's, it's, it's that when you help somebody and manage to reduce the noise to reduce the chaos in somebody's mind, that they, that when they become the greater degrees of health start returning to them, that they, they do rely far more, whether they know it or not, on probabilistic reasoning in their daily kind of functioning. And, and so I think there is a strong case to be made for a mathematical way of processing the world, it doesn't have to necessarily be conscious, made conscious to the person. It can be a bit overwhelming to, to try and impart that on somebody. But I think a brain that is not working properly, regardless of the degree to which there is chaos, that can be measured or quantifiably measured in that brain, somebody isn't, is tripping over themselves, because they, they've, they are unable to predict what might happen. And, and that is a terrifying prospect, because I think there is a, some reliance and some, some staunch dependability on the, the fact that the sun is going to rise and that it's going to set and that I can then focus on other things.
So that's just the, my comments more from kind of a personal front of being a patient and, and encountering the limitations of psychotherapy and what, what I would have, what I would have done in in those moments of need to have not, as you say, had a more, a deterministic view of what was happening.
And I've tried to transfer that or carry that over right, you're wrongly, into my own therapy as a, as a psychotherapist.
As, as, as, as, um, so a more personal thing of what it means to be, to feel chaotic and what I see, regardless of how confirmed it is, um, you know, or not.
Um, but, but chaos reigns and, and it's to try and not compound that, uh, is certainly something that, um, I'm interested in.
So, yeah.

Thank you, Bryn.
I'll also give my thought on what's at stake and what's on the table as a total non clinician, but what I saw in Alexi's paper and also what I see people coming towards.

I see the paper as Alexi making a, a real stable of concepts or portfolio of topics ranging from
the first principles of active inference, like surprise bounding to dynamic stochastic chaotic,
which are not the same thing, but they sometimes cohabitate.
So bringing a set of concepts into play, making it matter for the field so that it can matter for
the clinician so that it can matter for the patient, which is where you totally brought it home to Bryn.
And in my totally simplified model, I was seeing all this diagnostic machinery from personalized data,
EEG recordings, all these surveys and samples, all the different conversation therapy itself,
all these different measurements, inference.
And then it passes through the Markov blanket of the diagnosis.
And it's like, well, now we know that that person is X or they have X.
And then here's where we now branch back out with the modalities of, of treatment.
And so that bottlenecking has obvious relevance in terms of like getting a categorical grip on what
we basically all agree are wetware complex systems.
They're sensitive to initial conditions, all of that.
And what if though, like you brought up the example with the scar tissue, maybe it's literal
bodily scar tissue.
Maybe it's conceptual scar tissue.
What if that inference were able to be understood in the context of an attractor landscape?
And it's not just a singular attractor.
So we have multiple attractors.
We have a chaotic landscape.
And so what if we were able to perforate that blanket, speaking super loosely,
to bring in what we have already recognized as diagnostically relevant, like the fact that
patients who have differential entropy have this or that epileptic outcome or coma outcome,
bring in that machinery that led to the diagnosis, but now take it beyond the diagnosis and maybe even
beyond the clinician's concept directly into the hands of the patient.
Thank you so much.
Perhaps we can hear from Mark.
And yeah, I am not eager to talk.
I'll, if there is time, I'll say something.
But if not, that's okay.
Well, let me first of all say something about my screen.
You might, some of you have noticed that suddenly everything went black behind me.
That's because I have, I'm in South Africa where they have power problems.
And then a kind person in my household thought that they would help.
And they've put this huge spotlight here, which is shining in my eyes.
And so if I look as if I'm in a rabbit in headlights, it's because I am.
I sent a text message saying, please come and remove this big lamp you put here,

but they don't seem to have received the message.

Now back to our, our actual topic, to the extent that I'm able to not be distracted from it.

I think that the, when I asked Alexei what was at stake, his main answer was to do with a diagnosis.

Now, I don't mean that he's saying that that's all that's at stake, but I think he's saying this

illustrates what's at stake.

And so I want to, first of all, make sure that, because I'm not sure who our audience is and who's

going to be watching this in case anybody is, misunderstands.

I want to make clear that in psychoanalysis, we've long ago moved away from DSM and ICD type diagnosis.

We always thought that the nosology, the psychiatric nosology was wholly artificial and created divisions where they don't exist in nature.

And we replaced it with a far more dynamic approach to diagnosis.

And that's not just a matter of labeling.

It's a matter also of what kind of thought process lies behind the diagnosis.

In other words, conceptualizing what's going on here, rather than saying this is an instance of personality disorder and I'll specify which type.

You know, we don't think like that in psychoanalysis and really we never have.

In neuropsychoanalysis, we build in that same tradition.

In neuropsychoanalysis, I think that what Alexei was worrying about is he's saying, yeah, but you know, you use systems.

You say that there's this there's a certain number of drives and you use those to pass what you're seeing clinically.

And yes, we do.

And we have to because you can't just say what I'm seeing is chaos.

You know, what you've got to say is, well, what I'm seeing is chaos, but let me try my best to find the deeper structure of this.

And so what we do is much the same as what I was saying earlier about how brain and mind work.

Likewise, the clinician's mind works that way.

We have some deeper concepts that organize what we see.

And then more on the surface, we see the complexity.

You know, in other words, we don't think that we can reduce a patient just to one saying this patient has problems with this drive or something.

It's not as simple as that, of course.

But we do start with saying, well, these are the drives.

These are the attractors, to put it in terms of the attractor landscape that you're talking about.

These are the preferred states of the human phenotype, you know.

And in mental life, too, they are preferred states.

There are certain attractors that we're trying to stay in those, like, for example, fear.

We want to be safe.

We don't want to feel fear.

You know, with respect to attachment bonding, we want to stay close to our attachment figures.

We don't want to lose them.

With respect to rage, we don't want to be frustrated and have things standing in our way.

We want to be able to have free access to the objects of our needs and so on.

You know, those are these attractors.

And, you know, we have great difficulty staying in those homeostatic bounds.

And so, you know, it is a dynamic picture.

And then it's over the question of the different drives.

There's the highly individualized predictive model that each one of us comes up with in relation to those drives and in relation to the conflicts between those drives and so on.

So I think that this way of thinking that we use in psychoanalysis and neuropsychoanalysis is already of the kind that Alex is calling for.

So I think that what Daniel's just said is that therefore perhaps very important to make clear.

It's that he's offering us a language, an additional language, an additional, I don't just mean language, I should say, an additional set of concepts and a highly developed one, which we might find useful.

And so we should take this on board.

That sounds very convincing to me.

If that's what's at stake here, that Alex is saying this is a language that lends itself to the phenomena that you study and to the way in which you go about studying it, then it sounds to me as if Alex is right.

So that's the main thing I wanted to say in response to his comments in relation to what I said.

So if anyone else wants to say anything else, that's good with me.

But Alexi, if I've misunderstood and if there's something else that you think is at stake that I'm not getting, now's your chance to say it.

Alexi Koukowicz

Well, I just wanted to end the conversation when you said that if this language might be of any, even to consider this language in our future models, I'm a happy man.

There's nothing else I need to say.

And I just wanted to say that these are human things.

They're occupational hazards.

Some of the things I'm talking about, this paper I'm working on about narrative stuff, is that, you know, of course, you know, it is much easier for us to think in,

in categories.

Now, even in the psychoanalytic world, you know, what, you know, we do in the old, let's take the Macquillian system and the previous one.

We used to say three levels of functioning, you know, neurotic, borderline, psychotic.

These are categories.

They're not DSM categories.

They're not bipolar one, but they're categories nonetheless.

Then we have character styles, obsessive character style, histrionic character style.

They're categories.

Okay.

So to some degree, it's a human condition to put things in categories.

And I do think that this is a very strong historical reasons that we think this way.

We have generations of thinking this way in categories.

And yes, at the end of the day, when we paint a portrait of a patient, it's developed and it's like a tapestry.

All of these things weave into one another and it's complex.

But we start from basic categories.

And, you know, certainly if we consider that, you know, some things can be transient.

There can be emergence.

They're like that certain behavior in one context is a compulsion, but it's not a compulsion in another context.

And perhaps I'm saying trivial things to clinicians.

And also what, Breen, you said, just to comment, when you said, as a patient to reduce the noise and reduce the chaos, I detect the bias in the language.

You see how we call, you know, psychopathology in DSM disorders.

Disorder is a bad word for us.

And I'm trying to say chaos is healthy.

I spend most of my day in gamma functioning, which is a chaotic state.

It's not a bad thing.

So in the folk language, chaos is a bad word.

Disorder is a bad word.

I want to put things in order.

And then perhaps we're missing something.

What comes to mind is, I think, Mark, you had meetings with Michael Levin and Ian McGilchrist, who is, you know, big on talking about the mechanistic, you know, logical, linear, straight and narrow left hemisphere type.

Of course, it's not left hemisphere, but that's the model versus holistic, sensorial kind of right hemisphere artistic type.

And we're missing the second part.

Like Daniel's background is surrounded by art where, you know, we spend less time in the right hemisphere.

Maybe appropriately so.

But I think that there's some things that lend themselves to that level of understanding.

And we break things.

We look at psychopathology and human suffering like it's a carburetor sometimes where we, again, you know, put it in buckets.

That's my impression.

But what you said is wonderful.

And I, you know, I don't contradict with anything.

I agree with everything you said, Mark.

Alex, if I can just comment, you know, it's almost like an arc.

You know, the pot of gold at the end of the arc is the reduced chaos, the reduced noise in terms of the patients from the patient's perspective.

But how you get there, ironically, is actually by inducing states of tolerable chaos and then kind of reformulating that, reconsolidating that chaos.

So it's not that from, you know, healing or growth or whatever the patient might present to you for.

It has everything to do with minimal or marginal chaos induction and then, you know, reintegration of that.

So the whole process is premised on chaos, but in bite-sized doses, maybe one could say.

Right.

Right.

And just to finish up and then I'll stop talking.

Jonathan Shetler, who is, you know, all known and is, you know, widely respected as a psychoanalytic practitioner and teacher and writer.

He recently said that take a patient X with an analyst Y together, the diet, as emergent properties.

That is not the same thing as patient X with an analyst Z.

Right.

And show me one study where we take that into account.

We don't.

We just do a random, you know, clinical trial.

Then we do meta-analysis of random clinical trial.

What technique did the therapist use?

They used, you know, whatever four letter therapy they applied.

Nobody looks at the emergent properties in a diet, this matrix, you know, between the two.

And that's the language of dynamical systems.

This is Daniel Friedman, you know, ant colony.

You know, a single ant doesn't build bridges and grow mushrooms.

Ant colony does all those things.

We don't talk about emergence in psychoanalytic psychotherapy, do we?

It's true.

It's also a fundamental challenge.

After all, you can't replicate within one dyad or across dyads in a finite world.

So it does point to fundamentals.

In our last minutes, if each would like to give some thoughts, where do we go as we now be open of exponent away from each other in our paths?

So maybe Albert, Bryn, Mark, and then Alexei, what are your overall thoughts or reflections?

And then where are we heading?

Go for it, Albert.

Yeah, I was about to comment on Alexei about the emergent properties thing.

It's kind of the same point I made earlier.

And later, psychoanalytic theories do talk about emerging properties.

But yes, I think, as what Professor Solstice mentioned, this new set of concepts makes it very practical for us to communicate with each other.

Because psychoanalysis has a bad habit of reinventing new concepts every new decade.

So there's a lot of chaos, to use that word, within psychoanalytic discourse.

So yeah.

But overall, I thought I'm very grateful.

I think this was a very interesting talk.

So yeah, thank you.

Thank you.

Yes.

So by way of closing remarks, I agree with...

Actually, it seems to me that there's an abundance of agreement in this meeting.

I don't see any big controversy.

So that's nice.

But I think what Bryn said really captures it.

I liked it because what he said was so experienced near.

It was just describing what it feels like to be a patient and to be in a state of not knowing and uncertainty,

as opposed to a more stable kind of tranquility.

And then the ability to allow yourself to return to uncertainty.

If you've found answers that are too simple and that are too static to allow ourselves to go back into this state of not really knowing,

that I think really does describe what we try to do for our patients,

what we have to remember applies to ourselves, not only because we are ultimately patients too,

but because it applies to how the mind works.

It applies to science, how science works, you know, that we do need to find order.

But there is disorder out there.

And so we're always at risk of coming prematurely to a rigid system of belief.

And we have to return back to the coalface and back to the chaos.

But ultimately, where we want to be is back in that valley.

And so as much as Alexia is right to remind us that the highest levels of brain activity,

in many senses of the word, are entropic.

And that deep sleep and seizures, you know, are bad things.

As much as that's true, nevertheless, I think that the great trend of the mind,

it's one of its fundamental working principles is we're wanting to be more certain.

We want life to be more predictable.

We don't like to be in chaos.

And I think that that's an important thing to remember, you know,

while we are acknowledging the importance of chaos and of the highly unpredictable nature of the world

and of the phenomena we study in our field.

Nevertheless, we retreat from it and for very good reasons.

So those are my closing remarks.

And thank you.

And I'll be very brief.

Alexia, yeah, just thank you for the paper and for having such a strong focus on the patient

and on psychopath, well, on human, the human condition.

I think that's so nice to have that come out of the paper when at first glance it doesn't seem to be there.

So thanks for making it so human.

And then just Daniel for organizing and to spend time with Mark and Albert.

It's a great pleasure.

So thank you very much.

Alexi, final words?

I want to thank everybody and Mark and Bryn and Albert and Daniel and Andrea.

Thank you.

This is, I had a great fun and I'm very grateful, honestly, for your help and openness and flexibility.

One of the other reasons is kind of the time we live in.

Yuval Herrera made that comment that the rate of change is accelerating.

You know, when we were hunter-gatherers, uncertainty was norm.

You wake up, there's nothing in the fridge.

You have to go hunt and we tolerate an uncertainty better.

We could put our worldly possessions on the back.

And then we started living in the world of iPads.

iPhones were bombarded with information and we must minimize uncertainty.

We must, you know, have predictability in everything, which is not the natural world for mammals.

And now, you know, we live our life and then bang, COVID hits.

And then bang, you know, a certain person becomes the president of the United States, you know, and everything goes upside down.

So the black swans are hitting us more frequently.

So I think having the theory of that, you know, at our fingerprints and applying it to our work is probably, you know, beneficial.

But again, thank you all so much.

I really appreciate it.

Awesome.

Well, thank you all for joining.

The conversation continues.

So until next time.

Thank you.

Thank you.

Bye.

Bye-bye.

Thanks a lot.

Bye-bye.

Bye-bye.